

Question.1

```
import java.io.*;

class StudentGradeCalculator
{
    public static void main(String args[]) throws IOException
    {
        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

        int m1,m2,m3,m4,m5;

        int total;

        double percentage;

        String grade;

        System.out.print(" marks of Subject 1: ");
        m1=Integer.parseInt(br.readLine());

        System.out.print(" marks of Subject 2: ");
        m2=Integer.parseInt(br.readLine());

        System.out.print(" marks of Subject 3: ");
        m3=Integer.parseInt(br.readLine());

        System.out.print(" marks of Subject 4: ");
        m4=Integer.parseInt(br.readLine());

        System.out.print("marks of Subject 5: ");
        m5=Integer.parseInt(br.readLine());

        total=m1+m2+m3+m4+m5;

        percentage=(double)total/500*100;

        if(percentage>=90)
        {
            grade="A+";
        }

        else if(percentage>=80)
```

```
{  
    grade="A";  
}  
else if(percentage>=70)  
{  
    grade="B";  
}  
else if(percentage>=60)  
{  
    grade="C";  
}  
else if(percentage>=50)  
{  
    grade="D";  
}  
else  
{  
    grade="Fail";  
}  
  
System.out.println("\n RESULT ");  
System.out.println(" Marks : "+total);  
System.out.println(" percent : "+percentage+"%");  
System.out.println("Grade : "+grade);  
if(percentage>=50)  
{  
    System.out.println("Status : PASS");  
}  
else
```

```
{  
    System.out.println("Status : FAIL");  
}  
}  
}
```

Question 2

```
import java.io.*;  
  
class ElectricityBillGenerator  
{  
    public static void main(String args[]) throws IOException  
    {  
        BufferedReader br=new BufferedReader(new InputStreamReader(System.in));  
        String name;  
        int number,units;  
        double bill;  
        System.out.print(" Consumer Name : ");  
        name=br.readLine();  
        System.out.print(" Consumer Number : ");  
        number=Integer.parseInt(br.readLine());  
        System.out.print("Enter Units : ");  
        units=Integer.parseInt(br.readLine());  
        if(units<=100)  
        {
```

```
    bill=units*3;
}
else if(units<=200)
{
    bill=(100*3)+(units-100)*5;
}
else if(units<=500)
{
    bill=(100*3)+(100*5)+(units-200)*7;
}
else
{
    bill=(100*3)+(100*5)+(300*7)+(units-500)*10;
}
if(bill>5000)
{
    bill=bill+(bill*0.05);
}

    System.out.println("\nConsumer Details");
    System.out.println("Name : "+name);
    System.out.println("Consumer Number : "+number);
    System.out.println("Units Consumed : "+units);
    System.out.println("Total Bill : "+bill);
}
}
```

Question 3

```
import java.io.*;
```

```
class NumberAnalysisTool
```

```
{
```

```
    static boolean prime(int n)
```

```
    {
```

```
        if(n<=1)
```

```
            return false;
```

```
        for(int i=2;i<n;i++)
```

```
        {
```

```
            if(n%i==0)
```

```
                return false;
```

```
        }
```

```
        return true;
```

```
    }
```

```
    static boolean palindrome(int n)
```

```
    {
```

```
        int temp=n,rev=0;
```

```
        while(temp>0)
```

```
        {
```

```
            rev=rev*10+temp%10;
```

```
            temp=temp/10;
```

```
        }
```

```
    return rev==n;
}
```

```
static boolean armstrong(int n)
```

```
{
    int temp=n,sum=0;
```

```
    while(temp>0)
```

```
    {
        int r=temp%10;
        sum=sum+(r*r*r);
        temp=temp/10;
    }
```

```
    return sum==n;
}
```

```
static int sumDigits(int n)
```

```
{
    int sum=0;
```

```
    while(n>0)
```

```
    {
        sum=sum+n%10;
        n=n/10;
    }
```

```
    return sum;
```

```
}
```

```
static int reverse(int n)
```

```
{
```

```
int rev=0;
```

```
while(n>0)
```

```
{
```

```
rev=rev*10+n%10;
```

```
n=n/10;
```

```
}
```

```
return rev;
```

```
}
```

```
public static void main(String args[]) throws IOException
```

```
{
```

```
BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
```

```
int n,ch;
```

```
System.out.print("Enter Number : ");
```

```
n=Integer.parseInt(br.readLine());
```

```
do
```

```
{
```

```
System.out.println("\n1. Prime");
```

```
System.out.println("2. Palindrome");
```

```
System.out.println("3. Armstrong");  
System.out.println("4. Sum of Digits");  
System.out.println("5. Reverse Number");  
System.out.println("6. Exit");
```

```
System.out.print("Enter Choice : ");  
ch=Integer.parseInt(br.readLine());
```

```
switch(ch)  
{  
    case 1:  
        if(prime(n))  
            System.out.println("Prime Number");  
        else  
            System.out.println("Not Prime Number");  
        break;  
  
    case 2:  
        if(palindrome(n))  
            System.out.println("Palindrome Number");  
        else  
            System.out.println("Not Palindrome Number");  
        break;  
  
    case 3:  
        if(armstrong(n))  
            System.out.println("Armstrong Number");  
        else
```



```
System.out.println("Not Armstrong Number");
```

```
break;
```

```
case 4:
```

```
System.out.println("Sum of Digits : "+sumDigits(n));
```

```
break;
```

```
case 5:
```

```
System.out.println("Reverse Number : "+reverse(n));
```

```
break;
```

```
case 6:
```

```
System.out.println("Program End");
```

```
break;
```

```
default:
```

```
System.out.println("Invalid Choice");
```

```
}
```

```
}while(ch!=6);
```

```
}
```

```
}
```

Question 4

```
import java.io.*;
```

```
class ArrayStatistics
```

```
{
```

```
    public static void main(String args[]) throws IOException
```

```
    {
```

```
        BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
```

```
        int a[]=new int[10];
```

```
        int i,max,min,sum=0,even=0,odd=0,search;
```

```
        double average;
```

```
        boolean found=false;
```

```
        System.out.println("Enter 10 numbers : ");
```

```
        for(i=0;i<10;i++)
```

```
        {
```

```
            a[i]=Integer.parseInt(br.readLine());
```

```
        }
```

```
        max=a[0];
```

```
        min=a[0];
```

```
        for(i=0;i<10;i++)
```

```
        {
```

```
            sum=sum+a[i];
```

```
if(a[i]>max)
{
    max=a[i];
}
```

```
if(a[i]<min)
{
    min=a[i];
}
```

```
if(a[i]%2==0)
{
    even++;
}
else
{
    odd++;
}
}
```

```
average=(double)sum/10;
```

```
System.out.print("Enter number to search : ");
search=Integer.parseInt(br.readLine());
```

```
for(i=0;i<10;i++)
{
    if(a[i]==search)
```

```
{
    found=true;
    break;
}
}

System.out.println("\nMaximum Element : "+max);
System.out.println("Minimum Element : "+min);
System.out.println("Average : "+average);
System.out.println("Even Numbers : "+even);
System.out.println("Odd Numbers : "+odd);

if(found)
{
    System.out.println("Number Found");
}
else
{
    System.out.println("Number Not Found");
}

System.out.println("Array in Reverse Order : ");

for(i=9;i>=0;i--)
{
    System.out.print(a[i]+" ");
}
}
```

```
}
```

Question 5

```
import java.io.*;
```

```
class EmployeeSalaryManagement
```

```
{
```

```
    public static void main(String args[]) throws IOException
```

```
    {
```

```
        BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
```

```
        int n,i,max=0,min=0;
```

```
        int id[];
```

```
        String name[];
```

```
        double basic[],da[],hra[],pf[],gross[],net[];
```

```
        double average=0;
```

```
        System.out.print("Enter Number of Employees : ");
```

```
        n=Integer.parseInt(br.readLine());
```

```
        id=new int[n];
```

```
        name=new String[n];
```

```
        basic=new double[n];
```

```
        da=new double[n];
```

```
hra=new double[n];
pf=new double[n];
gross=new double[n];
net=new double[n];

for(i=0;i<n;i++)
{
    System.out.println("\nEmployee "+(i+1));

    System.out.print("Enter Employee ID : ");
    id[i]=Integer.parseInt(br.readLine());

    System.out.print("Enter Employee Name : ");
    name[i]=br.readLine();

    System.out.print("Enter Basic Salary : ");
    basic[i]=Double.parseDouble(br.readLine());

    da[i]=basic[i]*0.40;
    hra[i]=basic[i]*0.20;
    pf[i]=basic[i]*0.12;

    gross[i]=basic[i]+da[i]+hra[i];
    net[i]=gross[i]-pf[i];

    average=average+net[i];
}
```

```
for(i=1;i<n;i++)
```

```
{
```

```
    if(net[i]>net[max])
```

```
    {
```

```
        max=i;
```

```
    }
```

```
    if(net[i]<net[min])
```

```
    {
```

```
        min=i;
```

```
    }
```

```
}
```

```
average=average/n;
```

```
System.out.println("\nEmployee Details");
```

```
System.out.println("ID\tName\tBasic\tGross\tNet");
```

```
for(i=0;i<n;i++)
```

```
{
```

```
    System.out.println(id[i]+" \t"+name[i]+" \t"+basic[i]+" \t"+gross[i]+" \t"+net[i]);
```

```
}
```

```
System.out.println("\nHighest Paid Employee : "+name[max]);
```

```
System.out.println("Salary : "+net[max]);
```

```
System.out.println("\nLowest Paid Employee : "+name[min]);
```

```
System.out.println("Salary : "+net[min]);
```

```
System.out.println("\nAverage Salary : "+average);
```

```
}
```

```
}
```